

**Karnataka  
Physics- 2015**

**General Instructions:**

**All the parts are compulsory.**

**Answer without relevant diagram /figure/circuit wherever necessary will not carry any marks.**

**Direct answers to numerical problems without detailed solutions will not carry any marks.**

**PART-A**

**I. Answer all the followings:**

**1. Define S.I unit of charge.**

Sol: One coulomb is that charge which when placed at a distance of 1m from an identical charge in free space repels it and is repelled by a force of  $9 \times 10^9 N$

**2. A resistor is marked with colors red, red orange and gold. Write the value of its resistance.**

Sol: Resistance =  $22 \times 10^3 \pm 5$

**3. State Ampere's circuital law.**

Sol: Ampere's law:-

The line integral of magnetic field around any closed surface is equal to  $\mu_0$  times the net current in closed by the surface.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{net}$$

**4. What is magnetic declination?**

Sol: Magnetic declination at a place is the angle between meridians a geographic meridian at that place.

**5. Mention the significance of Lenz's law.**

Sol:

Lenz's law helps to find out the polarity of induced emf.

**6. How does the power of lens vary with its focal length?**

Sol:

Power of a lens varies inversely with focal length and is the reciprocal of focal length (in m)

$$\Rightarrow \text{Power} = \frac{1}{\text{focal length (in m)}}$$

**7. What is the conclusion of Davison and Germer experiment on the nature of electron?**

Sol:

Davison and Germer experiment concludes on confirms the wave nature of electron.

**8. Name the spectral series of hydrogen which lies in the ultraviolet region of electromagnetic spectrum.**

Sol: Lyman series lies in the ultraviolet region of electromagnetic spectrum.

**9. Define specific binding energy.**

Sol:

Binding energy per nucleon or a nucleus is known as specific binding energy

**10. What is attenuation in communication system?**

Sol: The loss of strength of a signal while propagating through a medium is known as attenuation in communication system.

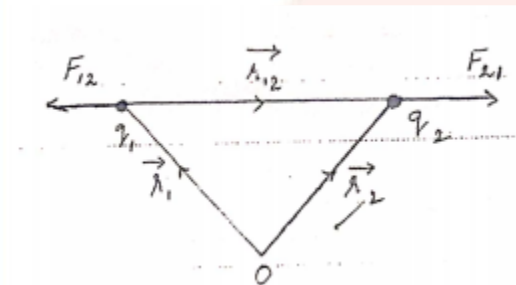
## PART-B

**II. Answer any five of the following question:**

**11. Write Coulomb's law in vector form. Explain the terms.**

Sol:

Coulomb's law in vector form:



Consider two point charges  $q_1$  and  $q_2$  with position vector  $\vec{r}_1$  and  $\vec{r}_2$  respectively.

Thus, the force on  $q_1$  due to  $q_2$  is

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}$$

Where  $\hat{r}_{12}$  is unit vector from  $q_1$  to  $q_2$

The force on  $q_2$  due to  $q_1$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

Where  $\hat{r}_{21}$  is the unit vector from  $q_2$  to  $q_1$ ,  $\epsilon_0$  is called absolute permittivity of force space.

**12. Mention two limitations of Ohm's law.**

Sol:

1. This law is valid to conductors only at constant temperature. The resistance of a conductor increases with the increase in its temperature. Hence, the V-I graph for a conductor would not be a straight line.
2. Insulators do not conduct electricity at all, so the Ohm's law is not applicable to the insulators in some cases.

**13. Write two properties of magnetic lines of force.**

Sol: Properties of magnetic lines of force:-

- 1) The magnetic lines of force due to a solenoid or a bar magnet form closed loop.
- 2) The tangent to the line of force never intersect each other. Because if they intersect, the direction of magnetic field at a point.
- 3) The magnetic lines of force never intersect each other. Because if they intersect, the direction of magnetic field would not be unique at the point of intersection.

**14. Current a coil falls from 2.5A in 0.0A in 0.1 second inducing an e.m.f of 200V. Calculate the value of self-inductance.**

Sol: Given that,

$$I_1 = 2.5A$$

$$I_2 = 0.0A$$

$$dt = 0.1\text{sec}$$

$$E = 200\text{V}$$

W.K.T

$$E = -L \frac{dI}{dt}$$

$$\therefore E = \frac{-L(I_2 - I_1)}{dt}$$

$$200 = \frac{-L(0.0 - 2.5)}{0.1}$$

$$200 = \frac{L \times 2.5}{0.1}$$

$$\therefore L = \frac{20}{2.5}$$

$$L = 8\text{Henry}$$

$$\therefore \text{self inductance} = 8\text{H}$$

**15. Mention two applications of infrared radiation.**

Sol:

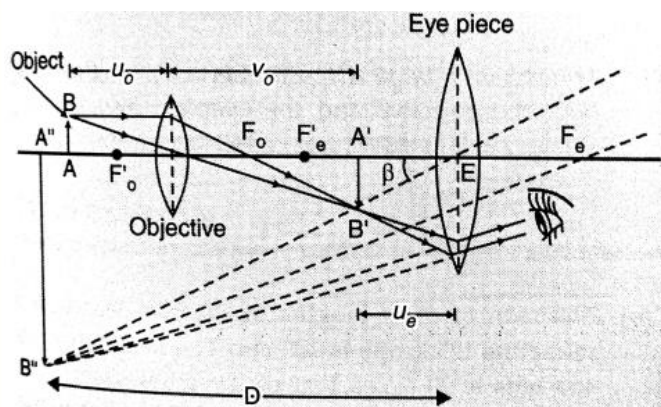
Application of infrared radiation:

Infrared lamps are used in physical therapy

Infrared radiations are used in solar cookers, solar water heaters etc.

**16. Draw the ray diagram of image formation in case of compound microscope.**

Sol:



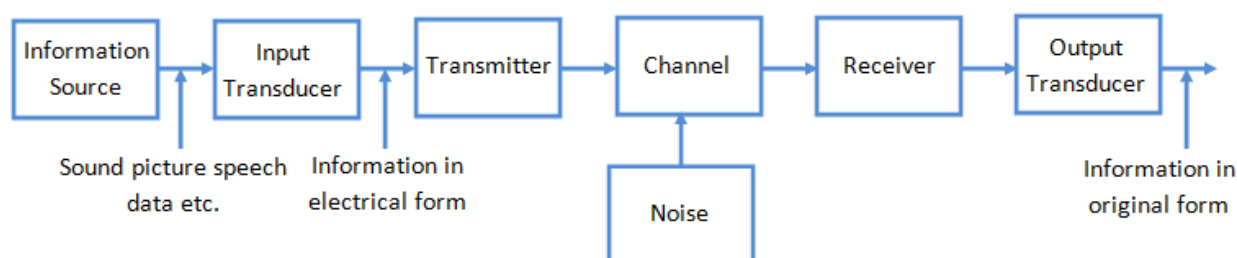
**17. What is photo diode? Mention its one use.**

A photo-diode is a semiconductor device consisting of a p-n junction made up of light sensitive material which is kept in reverse bias condition and when light falls on the junction, an electron is emitted because of inner photoelectric effect and a photo current is produced. This enables the diode to detect light. Thus, it is used as a light sensor.

**18. Draw the block diagram of generalized communication system.**

Sol;

Block diagram of general communication system.



### PART-C

III. Answer any five of the following question:

(5 x 3 =15)

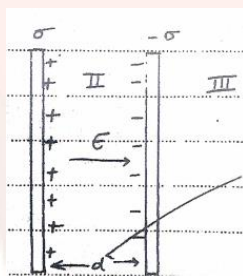
19. Derive the expression for capacitance of parallel plate capacitor.

Sol:

A parallel plate capacitor consists of two large parallel plane separated by a small distance.

Consider a parallel plate capacitor with area of each plate = A and distance between the plates = d let air be the medium between the plates.

Let the surface charge densities of the plates be  $\sigma$  and  $-\sigma$  respectively and the charge stored be Q and  $-Q$  respectively.



Since  $d^2 \ll A$ , we can use the result on electric field due to an infinite plate of uniform surface charge density.

In region I,

$$E = \frac{\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0} = 0$$

$\therefore$  E due to both plates as in opposite directions. In region II i.e, between the plates, the electric field due to both the plates is in the same direction and is directed from the positive plate to the negative plate.

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

Since the electric field is uniform,

$$\therefore V = E \times d$$

$$= \frac{\sigma}{\epsilon_0} \times d$$

$$V = \frac{Q}{A\epsilon_0} \times d \quad \therefore \sigma = \frac{Q}{A}$$

Since capacitance is the ratio of charge to potential.

$$\begin{aligned}\therefore C &= \frac{Q}{V} \\ &= \frac{Q}{\left(\frac{Q}{A\epsilon_0}\right) \times d} \\ C &= \frac{\epsilon_0 A}{d}\end{aligned}$$

If the area between plates is filled with a medium of dielectric constant  $k$ , then,

$$C = \frac{k \epsilon_0 A}{d}$$

**20. Explain with circuit diagram how to convert galvanometer into an ammeter.**

Sol: Since Galvanometer is a very sensitive instrument therefore it can't measure heavy currents. In order to convert a Galvanometer into an Ammeter, a very low resistance known as "shunt" resistance is connected in parallel to Galvanometer. Value of shunt is so adjusted that most of the current passes through the shunt.

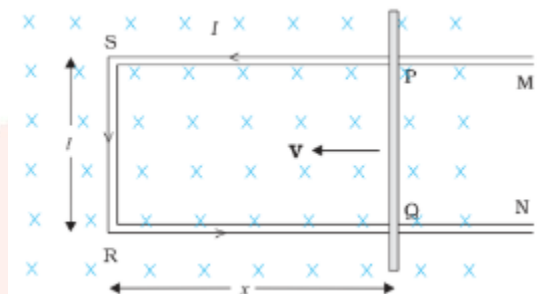
**21. Write three differences between diamagnetic and paramagnetic substances.**

Sol:

| Diamagnetic                                            | Paramagnetic                                                      |
|--------------------------------------------------------|-------------------------------------------------------------------|
| They are weakly repelled by an external magnetic field | They are durably attracted by an external magnetic field          |
| Susceptibility (magnetic) is small and negative.       | Magnetic susceptibility is small and positive.                    |
| Relative permeability is slightly less than one.       | Relative permeability is slightly greater than one.               |
| Susceptibility is independent of temperature           | Susceptibility is inversely proportional to absolute temperature. |
| Bismuth, lead etc.                                     |                                                                   |

## 22. Derive the expression for motional EMF induced in a conductor moving in a uniform magnetic field.

Sol: Let us consider a straight conductor moving in a uniform and time-independent magnetic field.



The figure shows a rectangular wire PQRS in which the conductor PQ is free to move.

Rod PQ is moved towards the left with a constant velocity  $v$ .

PQRS forms closed circuit enclosing an area that changes as PQ moves.

It is placed in a uniform magnetic field  $B$  perpendicular to the plane of our system.

Let the length  $RQ = x$  and  $RS = l$ ,

The magnetic flux  $\phi_B$  enclosed by the loop PQRS

$$\phi_B = Blx$$

As  $x$  is changing with time, the rate of change of flux  $\phi_B$  will induce an emf given according to Faraday's law of electromagnetic induction:

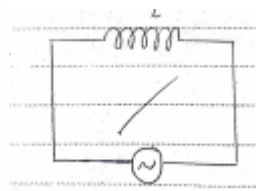
$$\epsilon = -\frac{d\phi_B}{dt} = -\frac{d(Blx)}{dt} = -Blv$$

This induced emf  $Blv$  is motional emf.

**23. Show that voltage leads current by  $\frac{\pi}{2}$ , when A.C voltage applied to pure inductance.**

Sol;

Consider on inductor of self-inductance  $L$  connected in series with an A.C source.



Let the applied sinusoidal voltage be given

$$v = v_m \sin \omega t \text{ --- (i)}$$

Where,

$v_m$  is the peak value of A.C voltage

$V$  is instantaneous value of A.C voltage

$\omega$  is angular frequency of source.

$t$  is time.

Applying Kirchhoff's to above circuit

$$\sum \epsilon = 0$$

$$\Rightarrow v - L \frac{dI}{dt}$$

Where  $-L \frac{dI}{dt}$  is the induced self-emf the inductor.

$$\text{i.e } v_m \sin \omega t = L \frac{dI}{dt}$$

Integrating on both sides

$$v_m \frac{(-\cos \omega t)}{\omega} = LI$$

$$I = \frac{-v_m}{\omega L} \cos \omega t$$

But,

$$-\cos wt = \sin\left(wt - \frac{\pi}{2}\right)$$

$$\therefore I = \frac{V_m}{W_L} \sin\left(wt - \frac{\pi}{2}\right) \text{----- (2)}$$

Comparing (1) and (2) we can say that voltage leads current by  $\frac{\pi}{2}$

Or, current lags behind voltage by  $\frac{\pi}{2}$ .

**24. What is interference? Write the condition for path difference in case of constructive and destructive interference.**

Sol:

Interference: when two light waves of same frequency and having constant phase difference travelling in the same direction superpose with each other, the intensity in the region of superposition, gets redistributed and at some points become minimum.

This phenomenon of redistribution of waves is called interference.

The graph shows the variation of intensity versus position in young's experiment of interference.

When both the slits are opened

**(figher)**

If we close one of the slits diffraction pattern is obtained.

**(figher)**

Condition for:

Path difference in case of constructive interference is given by

$$\text{Path difference } (\Delta x) = n\lambda$$

Where  $n \in N$

Condition for:

Path difference in case of destructive interference is given by

$$\text{Path difference } (\Delta x) = (2n-1)\frac{\lambda}{2}$$

Where  $n \in N$

**25. By assuming Bohr's postulates derive an expression for radius of  $(\Delta x) = n\lambda$  orbit of electron, revolving round the nucleus of hydrogen atom.**

**26. Distinguish between conductor and semiconductor on the basis of band theory of solids.**

**Sol:**

In the solids, the different energy levels of atoms overlap to form two types of bands c. valence band and conduction bands. The difference of energy between the bottom of conduction band and the top of valence band is called energy gap.

In case of conductors, valence band and conduction band overlap. Hence there is no energy gap. Therefore electron from valence band can move into the conduction band and become free for conduction. Thus, the conductivity is high and resistance is low.

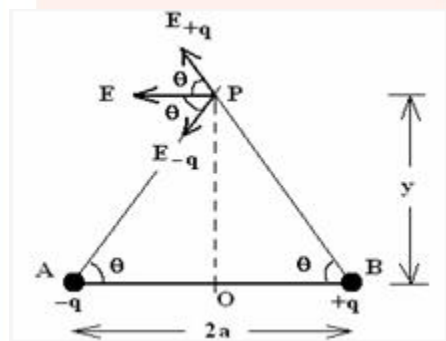
In case of semiconductor, there is a small energy gap. Between valence band and conduction band. Therefore only few electrons can cross the energy gap and become free electrons.

Thus, conductivity is lower than conductors but resistance is higher than conductors.

IV Answer any two of the following question:

**27. Derive an expression for electric field due electric dipole at a point on equatorial line.**

**Sol:**



Consider an electric dipole of dipole moment  $\vec{P} = q \times 2a \hat{p}$

Where  $q$  is the magnitude of either charge  $2a$  is the distance between the charges  $\hat{p}$  is the unit vector along the dipole axis.

Consider a point  $P$  at a distance ' $r$ ' from the center of the dipole on the equatorial line.

Then, the distance of point ' $P$ ' from both charges  $= (r^2 + a^2)^{\frac{1}{2}}$

The electric field  $\vec{E}$  due to  $+q$  at ' $P$ ' is

$$\vec{E}_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + a^2)} (-\hat{p})$$

The electric field  $\vec{E}_2$  due to  $-q$  at 'P' is

$$\vec{E}_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + a^2)} (-\hat{p})$$

$\vec{E}_1$  and  $\vec{E}_2$  are resolved into components parallel and perpendicular to the dipole axis

The perpendicular components cancel each other and the parallel components are added up. Thus, the net electric field at P is

$$\begin{aligned} E &= E_1 \cos \theta + E_2 \cos \theta \\ &= (E_1 + E_2) \cos \theta \\ &= 2E_1 \cos \theta \quad (\because E_1 = E_2) \end{aligned}$$

But,

$$\begin{aligned} \cos \theta &= \frac{a}{(r^2 + a^2)^{\frac{1}{2}}} \\ \therefore E &= 2 \left( \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + a^2)} \right) \frac{q}{(r^2 + a^2)^{\frac{1}{2}}} \\ &= \frac{1}{4\pi\epsilon_0} \frac{q \times 2a}{(r^2 + a^2)^{\frac{3}{2}}} \end{aligned}$$

But  $q \times 2a = p$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + a^2)^{\frac{3}{2}}}$$

In vector form  $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{P}{(r^2 + a^2)^{\frac{3}{2}}} \hat{n}$

Where  $\hat{n}$  is the unit vector opposite to  $\hat{p}$

Thus, the electric field is opposite to the direction of dipole moment.

If  $r \gg a$ , then  $E = \frac{1}{4\pi\epsilon_0} \frac{P}{r^3}$ .

**28. What is equivalent resistance? Derive the expression for effective resistance of two resistors connected in parallel.**

Sol:

Equivalent resistance is that single resistance which effectively replaces the combination of resistances such that the same current flows through it under the same potential.

Consider two resistances  $R_1$  and  $R_2$  connected in parallel as shown in the diagram. Two or more resistances are said to be connected in parallel if they are connected between two points such that on applying a potential difference across the combination, the potential difference is the same across each resistance but the current through the combination is equal to the sum of current through each resistance.

Let a battery of  $V$  volt be connected across the combination. If  $I_1$  and  $I_2$  be the current through  $R_1$  and  $R_2$  respectively. Then,

From Ohm's law,

$$V = I_1 R_1 \quad \therefore I_1 = \frac{V}{R_1}$$

$$V = I_2 R_2 \quad \therefore I_2 = \frac{V}{R_2}$$

But, by definition,

$$I = I_1 + I_2$$

$$I = \frac{V}{R_1} + \frac{V}{R_2}$$

$$I = V \left( \frac{1}{R_1} + \frac{1}{R_2} \right) \text{-----(1)}$$

Let the combination be replaced by a single resistance called equivalent resistance

Then,

$$I = \frac{V}{R_{eq}} \text{-----(2)}$$

From (1) and (2)

$$\frac{V}{R_{eq}} = V \left( \frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \quad \therefore R_{eff} = \frac{R_1 R_2}{R_1 + R_2}$$

This is effective resistance of two resistances connected in parallel.

## 29. Derive an expression for magnetic field strength at any point on the axis of a circular current loop using Biot-savart's law.

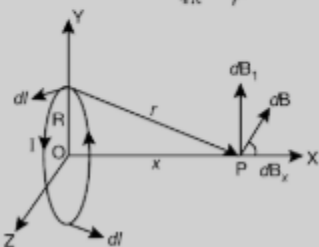
**Q.1.** Derive an expression for magnetic field strength at any point on the axis of a circular current loop using Biot-Savart's law.

[March 2015, 2014 M.Q.P., 2015, 2014]

**Ans.** Statement for Biot-Savart Law : The magnitude of magnetic field  $\vec{dB}$  due to current element is directly proportional to the current  $I$ , the elements length  $|dl|$  and inversely proportional to the square of the distance  $r$  of the field point. Its direction is perpendicular to the plane containing  $\vec{dl}$  and  $\vec{r}$ .

$$\vec{dB} \propto \frac{I dl \times \vec{r}}{r^3}$$

or 
$$\vec{dB} = \frac{\mu_0}{4\pi} \cdot \frac{I dl \times \vec{r}}{r^3}$$



The magnetic field due to  $\vec{dl}$  is given by Biot Savart law as

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I |dl \times \vec{r}|}{r^3}$$

Now  $dB_x = dB \cos \theta = \frac{\mu_0}{4\pi} \cdot \frac{I dl}{(x^2 + R^2)^{3/2}} \cos \theta$

$$= \frac{\mu_0}{4\pi} \cdot \frac{I dl}{(x^2 + R^2)^{3/2}} \cdot \frac{R}{(x^2 + R^2)^{1/2}} \quad 1$$

So,  $B_x = \int dB_x = \frac{\mu_0}{4\pi} \cdot \frac{IR}{(x^2 + R^2)^{3/2}} \int dl$

$$= \frac{\mu_0}{4\pi} \cdot \frac{IR}{(x^2 + R^2)^{3/2}} \cdot 2\pi R$$

$$= \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$$

(The y-components, of the field, add up to zero, due to symmetry)

$\therefore$  Magnetic field at P due to a circular loop

$$= B = B_x \hat{i} = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}} \hat{i} \quad 1$$

**Explanation :** A circular current loop produces magnetic field and its magnetic moment is the product of current and its area  $\vec{M} = IA \hat{n}$

[Scheme of Valuation, 2015]

## V. Answer any two of the following question:

### 30. Derive the expression for refractive index of the material of the prism in term of angle of the prism and angle of minimum deviation.

Sol:

Consider a prism made of glass of R.I  $n$  and refracting angle  $A$ .

Consider a principal section ABC of the prism. Let  $\hat{A}$  be the refracting angle.

(figher)

Let a ray of light PQ be incident on the prism at an angle of incidence  $i$ . It is refracted along QR and emerges along RS. In the absence of the prism, the say would have beveled along PQZ. But, due to the prism, it gets deviated by an angle of duration  $\delta$ . The normal drum the two surface meet at AN.

PQ and RS are produced backward to meet at M.

Let  $r_1$  the angle of refraction at surface ABC and  $r_2$  and 'e' be the angle of incidence and emergence at the surface AC.

From cyclic quadrilateral AQRN, we have

$$\hat{A} + \hat{QNR} = 180^\circ \text{ -----(1)}$$

From  $\triangle QMR$

$$\hat{r}_1 + \hat{r}_2 + \hat{QNR} = 180^\circ \text{ -----(2)}$$

From (1) and (2)

$$A = r_1 + r_2 \text{ -----(3)}$$

for  $\triangle QMR$

$\hat{SMZ}$  is the deflection

$$\hat{MQR} + \hat{MRQ} = \hat{SMZ}$$

$$i - r_1 + e - r_2 = \delta$$

$$\therefore \delta = i + e - (r_1 + r_2)$$

$$\therefore \delta = i + e - A$$

Thus as the angle of incidence increases, angle of deviation decreases and reaches a minimum value and then it starts increasing.

**(figher)**

From the graph, it is clear that for any angle of deviation  $\delta$ ,

There are two angles of incidence  $i$  and  $e$ .

When deviation angle is minimum then i.e

$$\therefore r_1 = r_2$$

Substituting these (3) and (4)

We get,

$$A = 2r$$

$$\therefore r = \frac{A}{2} \text{ -----(5)}$$

$$D = 2i - A$$

$$\therefore i = \frac{A + D}{2} \text{ -----(6)}$$

From Snell's law,

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

From (5) and (6)

This is the expression for R.I of the prism.

**31. Write Einstein's equation of photoelectric effect. Given Einstein's explanation of photoelectric effect.**

Sol:

Einstein's equation of photoelectric effect:

$$K_{\max} = h\nu - \phi$$

Where  $K_{\max}$  is the maximum kinetic energy photoelectrons.

$h$  is Planck's constant

$\nu$  is frequency of incident radiation

$\phi$  is work function

$\nu_0$  is threshold frequency

According to resisting photoelectric effect takes place due to the absorption of quanta of radiation called photons by the electrons of the metal.

If the energy of the photon is equal to work function of the photon is equal to work function of the actual, then the photoelectrons are just emitted from the metal surface.

If the energy of the photon is greater than work function, the difference in energy appears as the kinetic energy of the emitted photoelectrons.

a) From the equation, it is clear that the maximum kinetic energy of the emitted photoelectrons varies linearly with the frequency of incident radiation.

Intently is the number of quanta of radiations incident per unit area and therefore the maximum kinetic energy is independent of intensity as there is no effect of intensity.

b) As intensity increases, the no. of photons per unit area increases. Thus, more electrons absorb photons and are emitted. Thus, the photo current increases with intensity.

c) since the basic process incubated in photoelectric effect is the absorption of a single photon by a single electron therefore it is an instantaneous process even when the intensity is very low.

d) As K move must be non-negative  $h\nu > \phi$  or  $h\nu_0 \Rightarrow \nu > \nu_0$  therefore, there elicits a minimum frequency of incident radiation called there should frequency for a given metal below which no. photoelectric effect takes place.

$$V_0 = \frac{\phi}{h}$$

Where  $V_0 \rightarrow$  threshold frequency

$\phi \rightarrow$  work function of metal.

$h \rightarrow$  Planck's constant

**32. With a neat circuit diagram, explain the working of an n-p-n transistor in CE mode as an amplifier with input and output waveform.**

**VI Answer any three of the following question:**

**33. Two point charges +1C and -4C are 1m apart in air find the positions along the line joining the two charges at which resultant potential is zero.**

Given,

$$q_1 = +1nc$$

$$q_2 = -4nc$$

$$r^2 = 1m$$

i) Let 'x' be the distance of a point 'P' from  $q_1$  where the resultant potential is zero. Outside the line joining L charges. then,

$$V_1 + V_2 = 0$$

$$V_1 = -V_2$$

$$\frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} = -\frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$$

$$\Rightarrow \frac{1}{4\pi\epsilon_0} \frac{1}{x} = \frac{-(-4)}{1+x} \times \frac{1}{4\pi\epsilon_0}$$

$$\Rightarrow 1+x = 4x$$

$$\Rightarrow 3x = 1$$

$$\therefore x = \frac{1}{3}$$

ii) let 'x' be the distance of point 'P' from  $K_{\max}$  where the resultant potential is 'o' inside the line joining two charge. Then,

$$V_1 + V_2 = 0$$

$$V_1 = -V_2$$

$$\frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} = -\frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$$

$$\therefore \frac{q_1}{r_1} = -\frac{q_2}{r_2}$$

$$\Rightarrow \frac{1}{x} = \frac{-(-4)}{1-x}$$

$$\Rightarrow \frac{1}{x} = \frac{4}{1-x}$$

$$1-x = 4x$$

$$\Rightarrow 5x = 1$$

$$x = 0.2m$$

$\therefore$  The resultant potential is zero at 0.2m from  $q_1$  and at  $\frac{1}{3}m$  from  $q_2$ .

**34. Two cells of emf 2V and 4V and internal resistance  $1\Omega$  and  $2\Omega$  respectively are connected in parallel so as to send the current in the same direction through an external resistance of  $10\Omega$ . Find the potential difference across  $10\Omega$  resistor.**

Sol:

$$E_1 = 2V$$

$$E_2 = 4V$$

$$r_1 = 1\Omega$$

$$r_2 = 2\Omega$$

$$E_{\text{eff}} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$= \frac{2(2) + 4(1)}{1 + 2}$$

$$= \frac{8}{3}V$$

$$r_{eff} = \frac{r_1 r_2}{r_1 + r_2}$$

$$= \frac{1(2)}{1+2}$$

$$= \frac{2}{3} \Omega$$

$$\therefore R_{net} = r_{eff} + R$$

$$= \frac{2}{3} + 10$$

$$= \frac{32}{3} \Omega$$

$$I = \frac{E_{eff}}{R_{net}}$$

$$= \frac{8}{\frac{32}{3}}$$

$$= \frac{1}{4} A$$

$\therefore$  Potential difference across  $10\Omega$  resistance is

$$V = IR$$

$$= \frac{1}{4} \times 10$$

$$V = 2.5V$$

**35. A sinusoidal voltage of peak value 283 V and frequency 50Hz is applied to a series LCR circuit in which  $R = 3\Omega$ ,  $L = 25.48mH$  and  $C = 786\mu F$**

Find: (i) Impedance of the circuit

(ii) The phase difference between the voltage across the source and the current

(iii) The power factor.

Sol:

$$V_0 = \text{peak voltage} = 283V$$

$$f = 50Hz$$

$$L = 25.48mH$$

$$C = 786\mu F$$

$$X_L = \omega L$$

$$X_L = 2\pi fL$$

$$= 2(3.14)(50)(25.48) \times 10^{-3}$$

$$= 8000.72 \times 10^{-3}$$

$$X_L = 8.00072 \Omega \approx 8 \Omega$$

$$X_c = \frac{1}{\omega c}$$

$$= \frac{1}{2\pi fc}$$

$$= \frac{1}{2(3.14)(50)(786) \times 10^{-6}}$$

$$= \frac{1}{246804 \times 10^{-6}}$$

$$= \frac{10^6}{246804}$$

$$= \frac{1000000}{246804}$$

$$X_c = 4.052 \Omega$$

a.) Impedance = Z

$$z = \sqrt{R^2 + (X_c - X_L)^2}$$

$$= \sqrt{3^2 + (4.052 - 8)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$= 5 \Omega$$

$$\therefore Z \approx 5 \Omega$$

b.)

$$\tan \phi = \frac{X_L - X_c}{R} \quad \because X_L > X_c$$

$$= \frac{8 - 4.052}{3}$$

$$= \frac{3.948}{3}$$

$$\tan \phi = 1.316$$

$$\phi = \tan^{-1}(1.316)$$

$$= 52^\circ 46'$$

$r_1 + r$  Phase difference b/w source voltage and current  $r_1 + r$

c) Power factor  $\cos \phi = \frac{R}{Z}$

$$\cos \phi = \frac{R}{5}$$

$$\therefore \text{Power factor} = 0.6$$

**36. In a young's double slit experiment distance between the slits is 1mm. The fringe width is found to be 0.6mm. When the screen is moved through a distance of 0.25m away from the plane of the slit, the fringe width becomes 0.75mm. Find the wavelength of light used.**

Sol:

$$d = 1\text{mm}$$

$$\beta = 0.6\text{mm}$$

Let distance b/w slit screen = D

$$\text{Given, when } D = 0.25\text{m}, \beta = 0.6\text{mm}$$

But,

$$\text{Given, when } D = 0.25\text{m}, \beta' = 0.75\text{mm}$$

**37. Determine the mass of  $Na^{22}$  which has an activity of 5mCi. Half-life of  $Na^{22}$  is 2.6 years. Avogadro number =  $6.023 \times 10^{23}$**

Sol:

$$R = 5mCi$$

$$= 5 \times 10^{-3} \times 3.7 \times 10^{10} \text{ dis / s}$$

$$R = 18.5 \times 10^7 \text{ dis / s}$$

$$\text{Half life} = 2.6 \text{ years}$$

$$2.6 \times 3.5 \times 24 \times 3600 = 81993600 \text{ sec}$$

$$\text{Activity } R = \lambda N$$

But,

$$N = \frac{R}{\lambda}$$

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

$$= \frac{0.693}{2.6 \times 3.5 \times 24 \times 3600}$$

$$= \frac{0.693}{81993600 \text{ sec}}$$

$$\therefore N = \frac{R}{\lambda}$$

$$= \frac{18.5 \times 10^7 \text{ dis / s}}{\frac{0.693}{81993600 \text{ s}}}$$

$$= \frac{18.5 \times 10^7 \times 81993600}{0.693}$$

$$= \frac{15168815 \times 10^9}{0.693}$$

$$N = 21888623 \times 10^9$$

But,

$$N = \frac{\text{mass of Na}^{22}}{22 \times 10^{-3}} \times 6.023 \times 10^{23}$$

$$21888623 \times 10^9 = \frac{\text{mass of Na}^{22}}{22 \times 10^{-3}} \times 6.023 \times 10^{23}$$

$$\therefore \text{mass} = \frac{21888623 \times 22 \times 10^9 \times 10^{-3}}{6.023 \times 10^{23}}$$

$$= 79951801 \times 10^{-14} \times 10^{-3}$$

$$= 7.9951801 \times 10^{-7} \times 10^{-3}$$

$$= 7.9951801 \times 10^{-10} \text{ kg}$$

$$\therefore \text{Mass of Na}^{22} = 7.9951801 \times 10^{-10} \text{ kg}$$